

Week 8 Seminar Tasks

Matrices. Part 1

READING

Lecture 7 Notes (available on Blackboard)

Chapter 27. Croft, T and Davison R (2016) *Foundation maths*, 6th ed. Harlow: Pearson.

TASK 1. Matrix Order

See Lectures 7 Notes -Slides 3-4

Task 1.1

Establish *the order* of the following matrices:

$$A = (0, 0, 1, 1)$$

$$B = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 1 & 1 \\ 2 & 2 & 2 \end{pmatrix}$$

$$C = \begin{pmatrix} 1 & 1 \\ 2 & 2 \\ 3 & 3 \end{pmatrix}$$

$$D = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Task 1.2

How many elements would the matrices of the following orders have:

- A. 5×5 ;
- B. 1×10 ;
- C. 10×1 ;
- D. $n \times n$;

Task 1.3

In the table below, map the matrices of the sizes given in the left panel to the matrices drawn in the right panel. Which known types of matrices can you identify here?

- I. 1×4
- II. 2×2
- III. 5×1
- IV. 3×2
- V. 4×4

a) $(0, 0, 1, 1)$

b) $\begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}$

c) $\begin{pmatrix} 1 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ 3 & 2 & 1 & 0 \\ 4 & 3 & 2 & 1 \end{pmatrix}$

d) $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

e) $\begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{pmatrix}$

TASK 2. Identity Matrices

See Lectures 7 Notes -Slides 7-8

Provide a definition of the features of the *identity matrix*. Draw Identity **matrix A** of the size 3 and identity **matrix B** of the size 4.

TASK 3. Operations with Matrices

See Lectures 7 Notes -Slides 15-39

Task 3.1

- A. What are the conditions for adding matrices A and B?
- B. Calculate $\mathbf{A} + \mathbf{B}$ and $\mathbf{A} - \mathbf{B}$ for the following matrices; otherwise, explain why these operations are impossible.

1) $\mathbf{A} = \mathbf{B} = (0, 1, 2, 3, 4)$

2) $\mathbf{A} = (0, 1, 2, 3)$ $\mathbf{B} = (0, 1, 2, 3, 4)$

3) $\mathbf{A} = \mathbf{B} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

4) $\mathbf{A} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$ $\mathbf{B} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

5) $\mathbf{A} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$ $\mathbf{B} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$

6) $\mathbf{A} = \begin{pmatrix} 0 & 0 \end{pmatrix}$ $\mathbf{B} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$

7) $\mathbf{A} = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix}$ $\mathbf{B} = \begin{pmatrix} -1 & -4 \\ -2 & -5 \\ -3 & -6 \end{pmatrix}$

Task 3.2.

Multiply the following matrices A and B by *the scalar 2*.

$$\mathbf{A} = \begin{pmatrix} 1 & -4 \\ 2 & -5 \\ 3 & -6 \end{pmatrix} \quad \mathbf{B} = \begin{pmatrix} -1 & 4 \\ -2 & 5 \\ -3 & 6 \end{pmatrix}$$

Task 3.3

- A. What are the conditions for multiplying matrices A of the order $k \times l$ and B of the order $m \times n$?
- B. Where possible, multiply the following matrices; otherwise explain why multiplication is not possible
- a) Given $A = B = (1, 3)$, multiply A and B
 - b) Given $A = (1, 3)$ and $B = \begin{pmatrix} 2 & -3 \\ 3 & -2 \end{pmatrix}$ multiply A and B
 - c) Given $A = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$ and $B = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$ multiply A and B
 - d) Given $A = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$ and $B = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$ multiply A^T and B
 - e) Given $A = B = \begin{pmatrix} 2 & -3 \\ 3 & -2 \end{pmatrix}$ multiply A^T and B

Task 3.4

Let A be an *identity matrix of size 5*. Find what are

- a) A^2
- b) A^{10}
- c) A^{100}
- d) A^T
- e) A^{-1}

Task 3.5

For which of the following matrices can we compute A^2 and A^3 ? If the corresponding multiplication is possible, then compute the result; otherwise, give the reasoning why the computation is not possible

- i. A is of the size 4×5
- ii. A is a square matrix of size 2 with all elements 2
- iii. A is a row vector with 3 columns
- iv. A is a column vector with 6 rows
- v. A is a diagonal matrix where the main diagonal has 3 elements: 1, 2 and 3
- vi. A is a null matrix of the size 2×3

Task 3.6

Given a 3×3 matrix A, find A^2 by multiplying A by itself.

Matrix A:

$$A = \begin{bmatrix} 2 & 1 & 3 \\ 0 & 4 & -1 \\ 5 & 2 & 0 \end{bmatrix}$$

TASK 4. Matrix Equations

See Lectures 7 Notes -Slides 27-29

Task 4.1

Given that the following matrices are equal, find the values of p , q , r , and s .

Matrix C

$$C = \begin{bmatrix} 2p + 3 & 1 & \frac{r}{2} \\ \frac{5}{2} & q - 7 & 4 \\ 3p + s & 8 & 2r \end{bmatrix}$$

Matrix D

$$D = \begin{bmatrix} 7 & 1 & 5 \\ \frac{5}{2} & 6 & 4 \\ 9 & 8 & 20 \end{bmatrix}$$

Since the matrices are equal, each corresponding element in C and D must be equal. Set up equations for each element involving p , q , r , and s and solve them.

Task 4.2

For the matrices, A , B and C given below

$$A = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix}, B = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix}, C = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}$$

Calculate:

- a) Matrix D such that $2D = AB^T$
- b) Matrix D such that $5D = A^TC$

Task 4.3

For the matrices P , Q , and R given below:

$$P = \begin{pmatrix} 6 & 3 & 2 \\ 4 & 5 & 1 \end{pmatrix}, Q = \begin{pmatrix} 1 & 4 \\ 3 & 2 \\ 5 & 6 \end{pmatrix}, R = \begin{pmatrix} 10 & 8 \\ 7 & 14 \end{pmatrix}$$

Calculate:

Matrix S such that $3S = PQ - R$.